

CUSTOMER SHOPPING BEHAVIOUR

CCDS 321 - DATA MINING

Jou

Members

Lamees Alharthi
Joudi Abdulhadi
Layan Abdullah
Hala Alsubaie

Contributions

Task 1 & Task 3- Part 1
Task 2 & Task 3- Part 2
Task 2 & Task 3- Part 2
Task 1 & Task 3- Part 1

PROBLEM STATEMENT

Shopping businesses face challenges in understanding customer behaviour, payment preferences, and purchase frequency. This project aims to help businesses improve through data cleaning, analysis, and pattern recognition to predict customers' subscription behaviour based on Age, Gender, Shipping Type, and Item Purchased.

OBJECTIVES

- clean and preprocess the data before modeling.
- build a model that predicts whether the customer will subscribe or not.
- evaluate the model’s performance and interpret the results.

DATASET DESCRIPTION

This dataset describes customer shopping behavior, with 5,050 observations and 17 attributes, which contains 13 categorical and 4 numerical, covering customer information, purchase details, and behavioral attributes such as age, purchase amount, and subscription status.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
2	Age	Num	8	BEST12.	BEST32.
5	Category	Char	11	\$11.	\$11.
9	Color	Char	7	\$7.	\$7.
1	Customer ID	Num	8	BEST12.	BEST32.
14	Discount Applied	Char	3	\$3.	\$3.
17	Frequency of Purchases	Char	14	\$14.	\$14.
3	Gender	Char	6	\$6.	\$6.
4	Item Purchased	Char	10	\$10.	\$10.
7	Location	Char	14	\$14.	\$14.
16	Payment Method	Char	13	\$13.	\$13.
15	Previous Purchases	Num	8	BEST12.	BEST32.
6	Purchase Amount (USD)	Num	8	BEST12.	BEST32.
11	Review Rating	Num	8	BEST12.	BEST32.
10	Season	Char	6	\$6.	\$6.
13	Shipping Type	Char	14	\$14.	\$14.
8	Size	Char	2	\$2.	\$2.
12	Subscription Status	Char	3	\$3.	\$3.

DATA CLEANING

1. Missing Data Handling

- Method for numerical variables (Purchase Amount, Review Rating, Previous Purchases):
→ Mean imputation

```
proc stdize data=WORK.IMPORT
    out=WORK.FINAL_FIXED
    reponly
    method=mean ;
var 'Purchase Amount (USD)'n
    'Review Rating'n
    'Previous Purchases'n;
run;
```

- Method for categorical variable (Size):
→ Mode imputation (most frequent value = "M")

```
data WORK.FINAL_FIXED;
    set WORK.FINAL_FIXED;
    if missing(Size) then Size = "M";
run;
```

1. Duplicate Records

- Removed 50 duplicate rows
- Rows changed from 5,050 to 5,000

1. Inconsistent Values

- **Fixed repeated labels in Season and Frequency of Purchases**
- **Autumn → Fall**
- **Fortnightly → Bi-Weekly**
- **Every 3 Months → Quarterly**

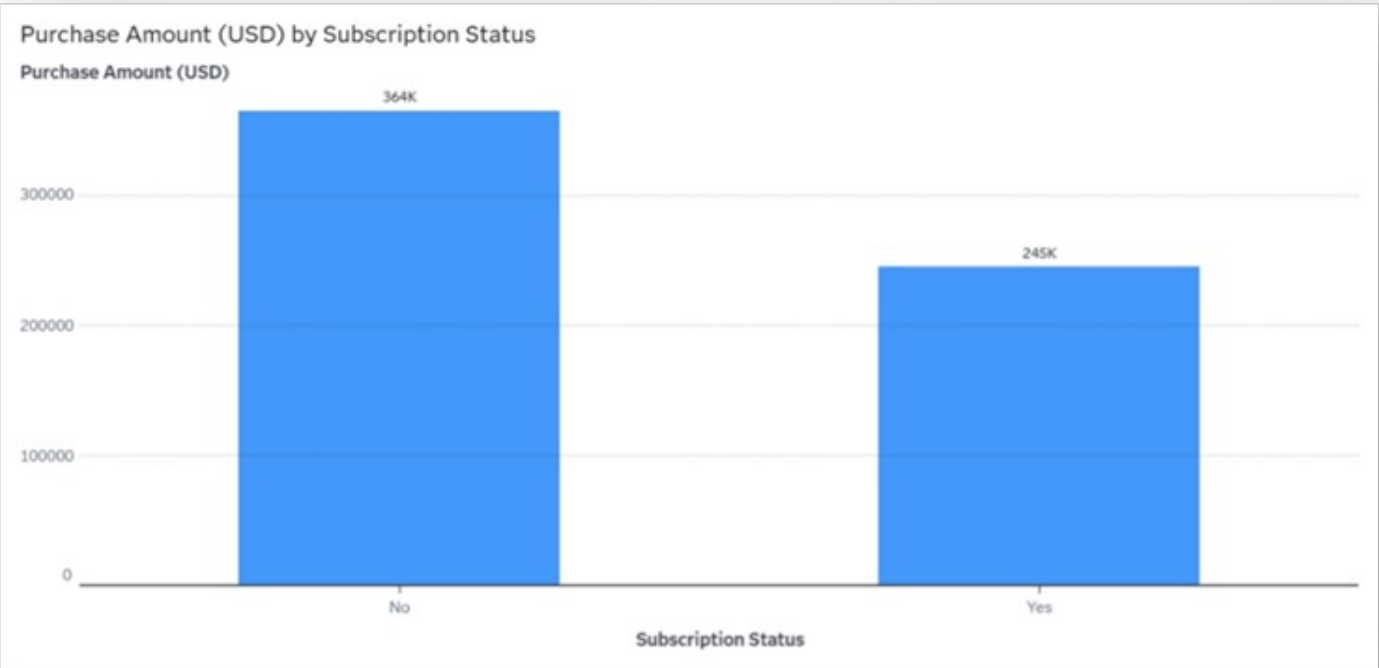
1. Outliers

- **Removed high Purchase Amount outliers**
- **Outliers were mainly in Shirt, Shoes, and Watch**
- **Final rows became 4,868**

EXPLORATORY DATA ANALYSIS

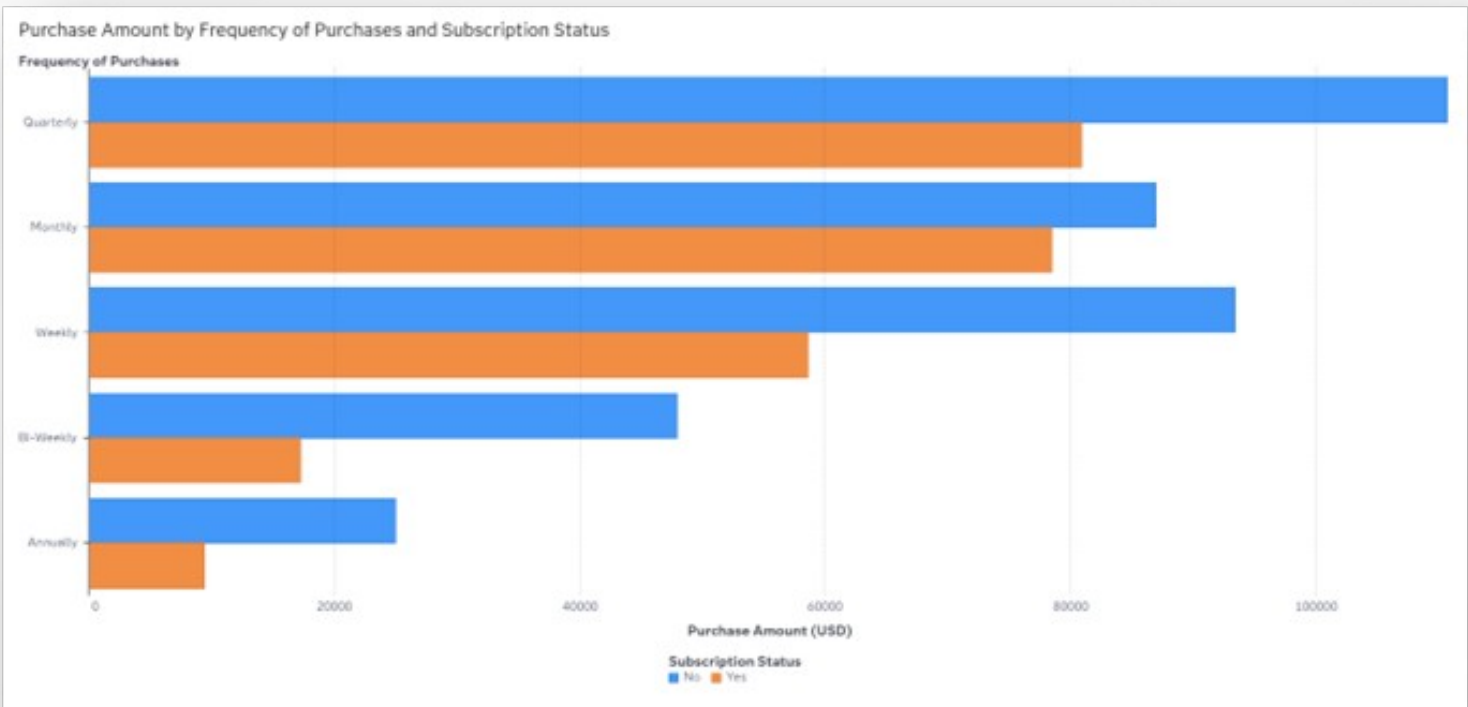
- BAR CHART ANALYSIS

Bar Chart Analysis The bar chart compares the total Purchase Amount (USD) between subscribers and non-subscribers.



- GROUPED BAR CHART

The bar chart shows the total Purchase Amount for each purchase frequency category, grouped by SubscriptionStatus.



EVALUATION APPROACH

Measure	Description	Definition of Value Function	Weight	Threshold
Accuracy	Overall proportion of correct predictions made by the model on the test partition	Accuracy Score = Accuracy	0.60	0.70
Simplicity	Measures tree complexity based on the size of the tree (number of leaves)	Simplicity Score = $1 - (\text{Leaves} - \text{Min Leaves}) / (\text{Max Leaves} - \text{Min Leaves})$	0.05	-
Lift	Measures the model's power at the 2nd decile compared to a random guess	Lift Score = $(\text{Lift Value} - \text{Min Lift}) / (\text{Max Lift} - \text{Min Lift})$	0.20	-
Stability	Measures the consistency of the DT performance between Training and Test sets for target event Subscription status	Stability Score = 1 if $\text{MR_Training} - \text{MR_Testing} \leq 0.05$, Otherwise 0	0.15	1

COMBINATION FUNCTION

Composite Performance Score Formula:

$$\text{Overall Score} = 0.60 \times V_{\text{Acc}} + 0.05 \times V_{\text{Simp}} + 0.20 \times V_{\text{Lift}} + 0.15 \times V_{\text{Stab}}$$

SUMMARY OF RESULTS

The decision trees were evaluated using performance measures such as Accuracy, Simplicity, Lift, and Stability. The Lift values show that Entropy and Gini have the highest Lift value and CHAID has a lower value. However, the DT7 and DT11 get the best overall score of 0.841.

LIFT CHART

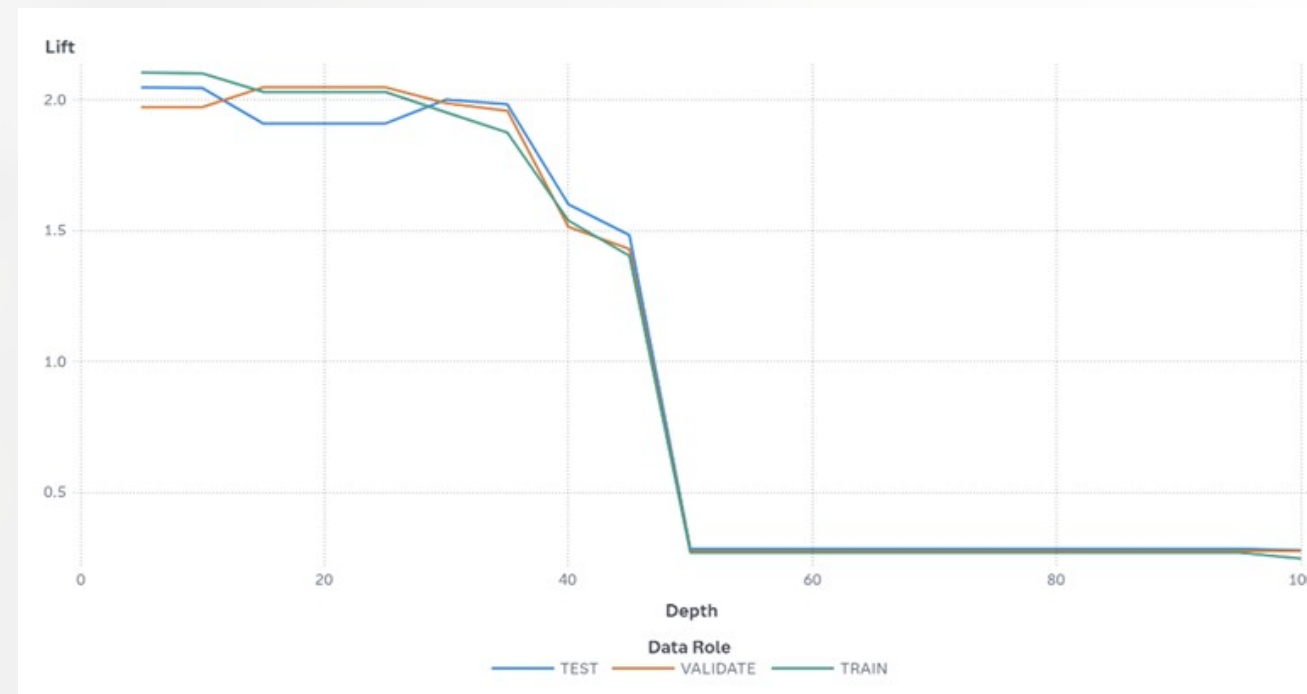


Figure 1 – Lift of Decision tree 7

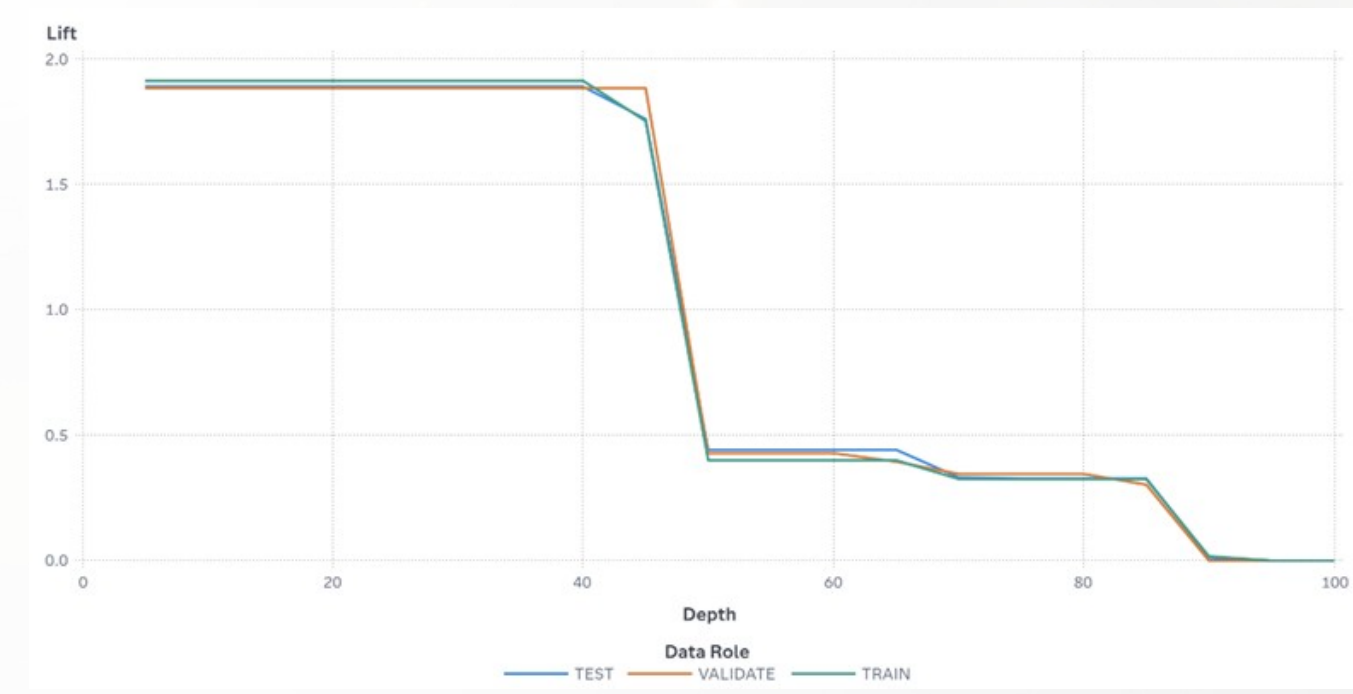


Figure 2 – Lift of Decision tree 1

The lift chart above compares DT1 and DT7 for the target event Subscription Status. At the second decile (20% depth), DT1 achieved a Lift of 1.91, while DT7 achieved a higher Lift of 2.1, meaning DT7 captured more subscribers than random selection. In addition, DT7 showed more consistent and stable Lift performance than DT1, indicating that DT7 is the more stable model.

DIFFERENT COMBINATION

Parameters:

- **Criterion: GINI**
- **Max Branches: 2**
- **Max Depth: 4**
- **Min Leaf Size: 100**

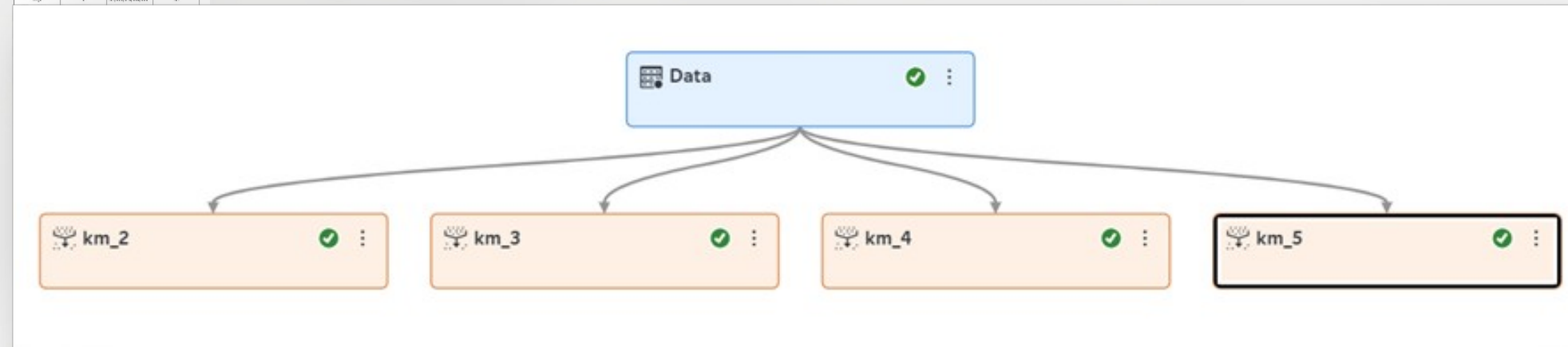
Results:

- **Accuracy = 0.7694**
- **Simplicity = 0.9688**
- **Lift = 1.494**
- **Stability = 1.0**
- **Overall Score: 0.860 → better than previous best (0.842)**

CUSTOMER SEGMENTATION

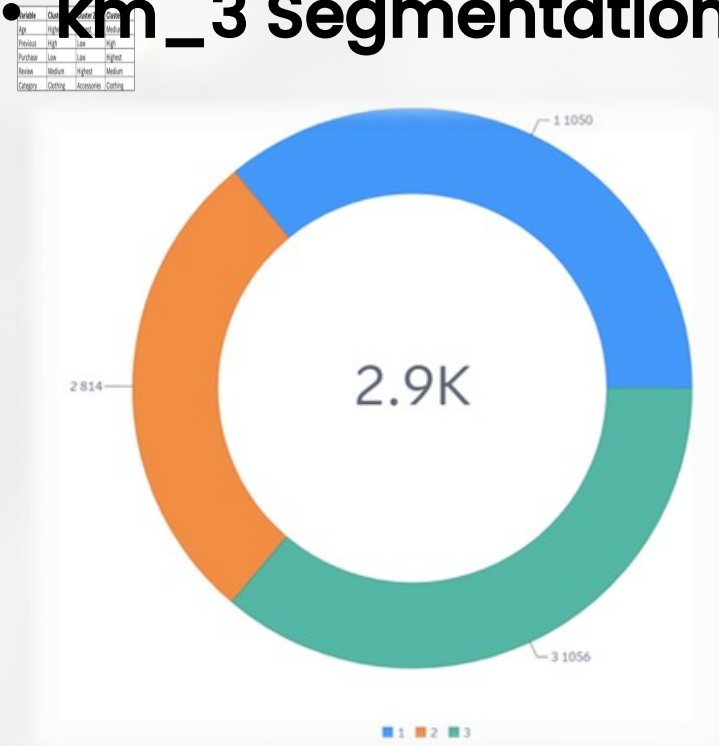
- **km_3 Segmentation:**

Segmentation Label	Number of Clusters	Cluster Size	Cluster	Out
km_2	2	1000, 1000	1000	
km_3	3	1000, 1000, 1000	1000	
km_4	4	1000, 1000, 1000, 1000	1000	
km_5	5	1000, 1000, 1000, 1000, 1000	1000	



CUSTOMER SEGMENTATION

- **km_3 Segmentation:**



Cluster	Size
Cluster 1	1050
Cluster 2	814
Cluster 3	1056

Cluster Centroids			
Cluster ID	Age	Previous Purchases	Purchase Amount (USD)
1	47.0615	26.3086	101.1439
2	41.7440	21.9602	103.2497
3	43.0417	26.5538	169.0896
Review Rating	Standard: Age	Standard: Previous Purchases	Standard: Purchase Amount (USD)
3.5809	0.1964	0.0839	-0.1124
3.8150	-0.1515	-0.2337	-0.1032
3.6709	-0.0666	0.1018	0.1844
Standard: Review Rating	Category	Discount Applied	Frequency of Purchases
-0.1231	Clothing	No	Quarterly
0.1651	Accessories	No	Quarterly
-0.0123	Clothing	Yes	Quarterly
Gender	Payment Method	Season	Shipping Type
Male	Debit Card	Spring	Express
Male	PayPal	Spring	Free Shipping
Male	Credit Card	Winter	Standard
Size			
M			
L			
M			

all clusters are more than 10% of the dataset, so there is no outlier cluster.
the smallest cluster is Cluster 2 with 814 customers, and it is greater than 10% of the dataset.

DECISION TREE COMPARISON

- Different DT models were compared using accuracy and simplicity
- Simplicity was measured using the number of leaves after pruning
- CHAID_2 and Gini_2 achieved the highest overall score (0.9680)
- Both models gave good accuracy with simpler tree structures
- Simpler trees performed better than deeper trees

BEST MODELS

- CHAID_2 → 0.9680
- Gini_2 → 0.9680

DT FOR CLUSTER ANALYSIS

- **Models tested: 8 (2 vs 4 branches, min leaf size = 244)**
- **Best models (Score = 0.8503):**
 - Entropy_2
 - Gini_2
 - InfoGain_2
 - InfoGain_4
- **Key findings:**
 - 2-branch models performed better than 4-branch models
 - CHAID models had Lift = 0 → worst performance

COMPARISON WITH TASK 2

- **In Task 2, all DT achieved the same accuracy of 0.9466. In Task 3, CHAID_2 and CHAID_4 achieved the highest accuracy of 0.7864. so, This shows that the DTs in Task 2 achieved higher accuracy than the DT from Task 3.**